

Swapan Kumar Sarkar

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[Swapan Kumar Sarkar | LinkedIn](#)

Portfolio link:

<https://swapancodes.github.io/website/>

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PROFILE SUMMARY

As Freelancer of RAG Engineer with experience in designing and deploying Gen AI applications using Large Language Models(LLMs) and Retrieval-Augmented Generation(RAG) architectures. Skilled in Python, LangChain, Embeddings , vector database(Milvus), semantic search, prompt engineering and API integration. Experienced in building document ingestion pipelines, embedding-based retrieval system and chatbot solutions. Passionate about leveraging AI technologies to solve real-world business problems and continuously improving model performance, retrieval accuracy, and user experience.

HIGHLIGHTS

- **Programming Languages:** Python
- **AI Techniques:** LangChain, Hugging Face, Milvus
- **Python Skills:** Variables, Data Types, Control Structures, Functions, works on Arithmetic, Relational & Logical operators, works on if statement, while loop & for loop, Attributes & Methods, Data Structure(List, Tuples, Dictionaries, Sets, Strings), List Comprehensions, Decorators, Generators, Iterators, Object Oriented Programming(Class, Objects, Inheritance, Encapsulation, Polymorphism), File Handling(Reading & Writing Files), Exception Handling(Try, Except), Functional Programming(Lambda Function, Map Function, Filter Function, Reduce Function), Multithreading, Pickling & Unpickling, Logging, Modules & Packages
- **PostgreSQL Skills:** data types, constraints & indexing(NOT NULL, PRIMARY KEY, FOREIGN KEY), operations(insert, update, delete), sorting, operators & wildcards(AND, OR, NOT, BETWEEN, LIKE), aggregate functions & grouping (COUNT, SUM, AVG, MIN, MAX, GROUP BY and HAVING), joining data(INNER JOIN, LEFT JOIN, RIGHT JOIN), subqueries, transactions(commit & rollback), Views, Stored Procedure, Window Functions, CTE(Common Table Expression), Triggers
- **Frameworks:** Skills in developing & integrating models using FastAPI.
- **Front-end Skills:** Basic Knowledge of (HTML, CSS, JavaScript)
- **Git & GitHub :** working knowledge of version control system on (e.g.,Git).
- **Machine learning Skills:** Missing value handling, Encoding, Outlier handling, Feature scaling, Feature selection technic, Feature extraction(PCA), Regularization, Confusion matrix, Cross validation, Bagging technic
- **Machine learning models:** Multiple linear regression, Polynomial regression, Logistic regression, Naïve bayes, Decision trees, SVM, KNN, Random forests, k-means clustering, Hierarchical clustering, DBSCAN clustering, Apriori algorithm, FP growth algorithm
- **MachineLearning Frameworks:** TensorFlow, Keras
- **Deep learning models:** ANN,CNN,RNN
- **Tools and Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV

WORK EXPERIENCE

RAG Engineer, Intelligent.ai as freelance basis (June 2024 – Present, WB)

Roles & Responsibilities:

- Developed and maintained Retrieval-Augmented Generation(RAG) applications using LLMs to improve knowledge retrieval and question-answering capabilities.
- Built document ingestion pipelines for processing PDFs, Word documents, web content, structured data sources.
- Implemented semantic search solutions using embedding and vector database Milvus.
- Designed and optimized chunking, embedding and retrieval strategies to enhance search relevance and reduce LLM hallucinations.
- Developed AI-powered chatbot.
- Integrated Large Language Models through APIs for content generation, summarization.

Projects:

- PDF RAG chatbot link: <https://ragpipe.streamlit.app>

Sr. Network Engineer, Nipun Net Solution Pvt Ltd (June 2023 – June 2024, Hyderabad)

Roles & Responsibilities:

- Cisco Switch 2960 Configuration(Vlan, Trunking,)
- Cisco Switch 3850 Configuration (Vlan, LACP Protocol, ip routing as per vlan wise,telnet configure, ip helper-address configure)
- configuration Scope of DHCP Server.
- Aruba 2930 Switch Configuration (Vlan, Tagged,Untagged,Lacp)
- Switch Configure Backup
- HSRP configuration
- 802.1x port configure and troubleshooting in switchport
- MAC binding in switchport

Technical Support Engineer, Informatics Technologies Pvt Ltd (Aug2022 – May2023, Ahmedabad)

Roles & Responsibilities:

- I have worked on Python project to build one Pharmacy Management System Software written Python code and established connectivity with MySQL Database. Responsible for writing reusable, testable and efficient code ensuring the applications performance, security and scalability

Desktop Support Engineer, IMSI Staffing(A TeamLease Company) (Feb2021 – Aug2022 , Haldia,WB)

Roles & Responsibilities:

- All Require Software Installation and Troubleshooting of Customer Oriented Problems
- System Image Backup & Recovery
- Computer join in domain
- Port number of POP3, SMTP, IMAP
- Files sharing
- installation of Windows server 2012 R2 (PDC & ADC) both
- install Roll & Features of ADDS, DNS, DHCP
- create User,Group & Computer Accounts in Active Directory
- create Group Policy Objects(Wallpapers, Pendrive access block, password policy etc)
- configuration Scope of DHCP Server
- DHCP Server Backup & Restore
- Full Server Backup
- add Forward Lookup Zone
- add Reverse Lookup Zone
- Knowledge of ADFS and LDAP
- Clear Concept how office 365 integrate with Active Directory
- knowledge about Microsoft Intune and OU
- Hikvision Bullet Camera install, Configuration & Troubleshooting

- NVR Configuration and troubleshooting
- iVMS-4200 install and configuration
- Camera recording download

System Engineer, Precision Techserve (M) Pvt. Ltd (Oct2019 – Jan2021 , Hooghly, WB)

Roles & Responsibilities:

- Installation and Configuration of Windows based Operating Systems (windows 10&11)
- Software Installation and Troubleshooting of all types of Customer Oriented Problems
- Configure and Troubleshooting of Desktop & laptop for L1 & L2 Backup and Recovery
- Network Settings for TCP/IP and Internet Setup
- Configure and Troubleshooting of Printer, Scanner & LAN Related Issue reported by users
- Installation and Configuration of Wireless Data Cards in users laptops
- Active Directory User creation and deletion
- Maintenance and Troubleshooting of connectivity problems using ping, traceroute
- Users Group Policy creation and modification
- IBM Lotus Notes Configuration

Software Support Engineer, Aidni Infotech Pvt Ltd (April 2019 – Oct2019 , Kolkata)

Roles & Responsibilities: Web Development , back-end and front-end software support.
Discuss with clients to understand the requirements.

Private Tutor (Mathematics), Self-Employed (2014 – 2019, WB)

Roles & Responsibilities: Works as a private Tutor of Class IX & X on Mathematics Subject of CBSE & WBBSE Board.

QUALIFICATION

- **B.Sc. in Mathematics, University of Calcutta, 2014**
- **Higher Secondary (Science), W.B.C.H.S.E, 2011**

TECHNICAL QUALIFICATION

- **Post Graduate Diploma in Software Engineering, from THE INSTITUTE OF COMPUTER ENGINEERS [INDIA]**
- Completed a Networking Course **CCNA** on Routing & Switching from IINT During Mar-2017 to Feb-2018

SOFTSKILLS

- Problem-Solving
- Communication
- Collaboration
- Time Management
- Critical Thinking
- Adaptability

PERSONALDETAILS

Date of Birth: February 3, 1992
Languages Known: Bengali, English, Hindi
Nationality: Indian
Marital Status: Unmarried
Gender: Male
Category: General